

WORK EXPERIENCE AND RESEARCH

Research Assistant

Apr. 2024 - Present

Nepal Applied Mathematics and Informatics Institute for research

Lalitpur, Nepal

Supervisor: Dr. Binod Bhattarai

- Working on research projects focused on **trustworthy machine learning**.
- Developed an **OOD detection method** for pathology segmentation in volumetric medical images, specifically MRI and CT scans, based on **self-supervised learning** with synthetic anomalies, leading to a **5th-rank** in the MICCAI 2024 challenge.

Undergraduate Research Intern

Aug. 2023 - Mar. 2024

Nepal Applied Mathematics and Informatics Institute for research

Lalitpur, Nepal

Supervisor: Dr. Binod Bhattarai

- Worked on an **Active Learning framework** to determine the most effective **uncertainty-based sampling technique** to address dataset imbalance in a multi-class object detection task.
- Trained deep learning models and conducted an ablation study for detecting gait abnormalities in warehouse environments.

EDUCATION

Pulchowk Campus - Institute of Engineering, Tribhuvan University

Nov. 2019- Apr. 2024

B.E in Electronics, Communication and Information

- Achieved Rank 104 in the university entrance examination out of nearly 18,000 candidates - **top 0.6%**, and received the scholarship provided by the Government of Nepal covering full tuition expenses
- Graduated with **75.73%**, ranked in the top 5% of students, and as the **top-performing female student** in the program.

PUBLICATIONS

- Chhetri, A.**, Korhonen, J., Gyawali, P., & Bhattarai, B. (2025). **NERO: Explainable Out-of-Distribution Detection with Neuron-Level Relevance in Gastrointestinal Imaging**. *In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2025*

SELECTED ACADEMIC PROJECTS

Improving Adversarial Image Robustness For Preventing Image Manipulation

Oct. 2023 - Apr. 2024

Capstone Project

[Link to Repo](#)

- Developed a **JPEG-resistant image immunization pipeline** to prevent text-guided AI manipulation, specifically for text-guided diffusion models.
- Investigated the impact of text prompts on the **success of adversarial attacks** against **text-guided diffusion models**.

Real-Time Visual Element Generation Using GANs

Dec. 2022 - Apr. 2023

Academic Project

[Paper](#) | [Link to Repo](#)

- Developed a **real-time neural rendering system** for generating realistic grass in video games.
- Experimented with existing architectures, performed result analysis, and designed new architectures that achieved both visual fidelity and low inference time.

TEACHING EXPERIENCE

Fifth Annual Nepal AI School - NAAMII

27th Dec., 2024 - 6th Jan., 2025

Teaching Assistant

[Certificate](#)

- Performed live demonstrations of adversarial attacks in image models and LLMs to an audience of 100 students under the supervision Prof. Yash Raj Shrestha (University of Lausanne, Switzerland).
- Assisted in lab sessions on Geometric Deep Learning, Active Learning, and Self-Supervised Learning.

AWARDS AND HONORS

Nepal-US Hackathon - Category Winner (Innovative) Nepali Leaders Network	March 2026
Eastern European Machine Learning - Full Scholarship ANNT, Google DeepMind, University of Sarajevo	July 2025
Mentee, Women Leaders Mentorship Programme, <i>EU in Nepal</i> Mentor: H.E. Veronique Lorenzo , EU Ambassador to Nepal, 4% acceptance rate	Dec. 2024–October 2025
Medical OOD Analysis Challenge, 5th Place , Pixel-level Task MICCAI 2024	October 2024
Hacking for Humanity 2023, 2nd Runner Up Sheqal Foundation, Nepal's largest all female hackathon	February 2023
GritFeat AI Health Hackathon, 1st Position GritFeat Solutions	January 2023
ABU-Robocon 2022, 2nd Runner Up Asia-Pacific Robot Contest	August 2022
3rd Annual Nepal AI School - Full Scholarship Nepal Applied Mathematics and Informatics Institute for Research	December 2021
Mahatma Gandhi Scholarship Embassy of India	2017-2018

VOLUNTEER EXPERIENCE

- **Organizing Committee** Oct. - Dec., 2024
Fifth Annual Nepal AI School - NAAMII
 - **Lead Organizer** for the AI School Hackathon, where I managed logistics, including stakeholder coordination, task division, and event planning both prior to and during the event.
 - **Served on the admissions committee** and evaluated around 150 applications under scholarship category.
- **AI Trainer, Right to Information** 10th Aug. - 14th Aug., 2024
Sancharika Samuha, The Asia Foundation, USAID Nepal
 - Led an AI training session in eastern Nepal with **50 journalists from 6 districts**, focusing on enhancing their understanding of AI technology.
 - Provided foundational AI knowledge and conducted hands-on training, demonstrating how tools like **ChatGPT, DALL-E, and Claude** can be used by journalists in Nepal to enhance productivity.
- **Children in Technology, 2023** 26th - 29th Nov., 2023
LOCUS 2023
 - Led a team of **13 undergraduate students** to **5 underserved rural communities** in eastern Nepal, engaging with **13 schools** and approximately **2,000 students** on internet safety and cyber law.

AFFILIATIONS

- **Event Manager** Dec. 2021 - Feb. 2023
IT Club, Pulchowk Campus
 - Organized **5 events**, each attended by approximately **40 students**, covering topics such as low-level system programming and UI/UX design.
 - Oversaw **technical setups** and **logistics** for all club events.
- **Student Member** Dec. 2019 - Aug. 2022
Robotics Club - Pulchowk Campus
 - Worked on multi-view geometry and object localization using stereo vision for a ball-throwing robot in **ABU-Robocon 2022**.
 - Organized a **robotics training camp** for 35 primary school students at 2 public schools, aiming to encourage early interest in robotics and STEM.
 - Mentored junior members for **ABU-Robocon 2023**.